

PRO-VET

Proteomics Visualization and Evaluation Tool

Installation:

Pro-Vet (Pro-VET_ver_00.40.exe) should be downloaded with two .pdf files and one image file. The two .pdf files are i) this instructions file (Pro-VET_instructions.pdf) and ii) file describing the peptide properties available for the AAindex and their index numbers (AA_index_details.pdf). In addition, an image file (PRO-VET_image.jpg) is included. All four files should be put in a single folder and the resulting results folder and subfolders will be created in this folder. Only the .exe file is needed for most operations.

Steps to Use:

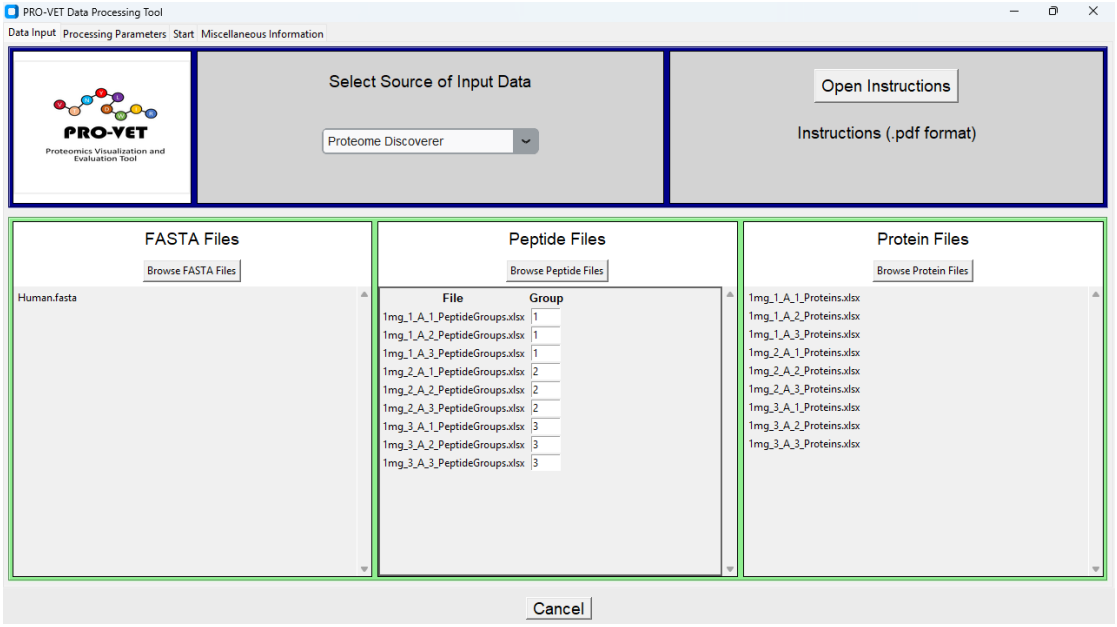
1) **Generation of Data**
   This program is designed to work with proteomics data generated using DDA workflows. Search instrument generated "raw" files with your favorite search engine. Each sample/run should be searched independently. All searches should be performed using the same search engine, search parameters and fasta database(s). Results from several of the more common search engines/proteomics analysis software suites can be directly imported into Pro-VET (see Appendix 1).
2) **Data Import.**

Prepare data for import into Pro_VET. For any software, one can manually convert the results into .csv files and import into this software. For each run, two files are needed. The first is the peptide file which must contain at least 4 columns, with the headings "seq", "accession", "PSM", and "Area". Sequence ("seq") is the peptide sequence using all upper-case 1-letter abbreviations for the 20 canonical amino acids. Peptide spectral matches ("PSM") or spectral count are the number of times that peptide is matched to a spectra. Accession ("accession") is the accession number of the protein that matches that peptide sequence. This must be the same accession number used in the fasta database in the search and there should only one per peptide. While certain peptides can be derived from more than one protein, its most likely associated protein should be used. This is termed differently for different search engines. Area ("Area") is the peak area for that peptide. If only peak heights are known, these can be used in place of the areas but the heading should remain "Area". For If no peak areas are available, simply put 1.0 into all entries of that column. (Do not select "area" or "log of area" options during processing but most functions will work properly). An optional fifth column for modifications, entitled 'mods', can be used. The current version of Pro-VET only examines Met oxidation and should use the format such as "1xOxidation [M5]". This indicates that one Met at position 5 is oxidized. This file should be saved as a .csv (Comma-Separated Values) file with file name underscore ('_') then "_peptides.csv". For example, "sample_1A_peptides.csv". The second file should be the protein file with at least 4 columns with the heading "description", "accession", "PSM", and "prot_len". The description ("description") generally includes that protein name and is how the protein is described. Accession number ("accession") number and "PSM" are as described above. The protein length ("prot_len") should also be included. These files should also be .csv files, labelled with sample name, followed by underscore and "_proteins.csv". For example, the protein file corresponding to the peptide file above should be "sample_1a_proteins.csv". Several manually created files should be available with the distribution of this software, entitled: "manual_input_example_1_peptides.csv", and "manual_input_example_1_proteins.csv" for sample 1. Similar files for example 2 and 3 should also be available. All these column and file names are case sensitive and do NOT include the quotation marks. These files should be relatively easy to generate in Excel for those familiar with that software and then exported as a .csv file.

In addition to these protein and peptide files, a fasta file used for these searches with a ".fasta" extension is needed. This should be identical to the one used during the searches and in the format from uniport.org. More than 1 fasta file can be used but it must match the fasta file(s) used during the searches.

If the various samples are associated with groups, you can assign each sample to a group. If groups are used, a minimum of 3 samples per group is recommended.

The data import page should appear similar to below with your data in the associated fields:



3) The General Analysis parameters must be selected. These are:

a. <u>Select Processing Method.</u>

- **Single or Multiple Protein Analysis.** Indicate if you want analysis of a single protein or multiple proteins.
- **Calculation Method.** For some analysis, one of several calculation methods are used. The four options are PSM, log 2 of PSM, peak area (or peak height) and log 2 of peak area (or peak height). These are used for calculating similarity index and displaying similarity values.
- **Base Run.** (Optional). Base run, or reference sample, is the sample that is used to compare all others. It will appear first in most analysis.
- **Testing Only.** Do NOT check. Only for testing.

b. <u>Parameters for Single Protein Analysis. (Only used for single protein Analysis.)</u>

- **Protein Selection Method.** For single protein analysis, there are several options available. Protein can be selected randomly from one of the top 500 (???) most abundant proteins based on spectral count. They can be selected by user, which requires user to put protein accession number in entry box. They can also be selected by rank. Rank is the rank of the protein based on the number of PSMs for all samples.

The most abundant protein is 1 and they are in decreasing order. This also requires user to specify the rank in the proper entry box. (10 is default).

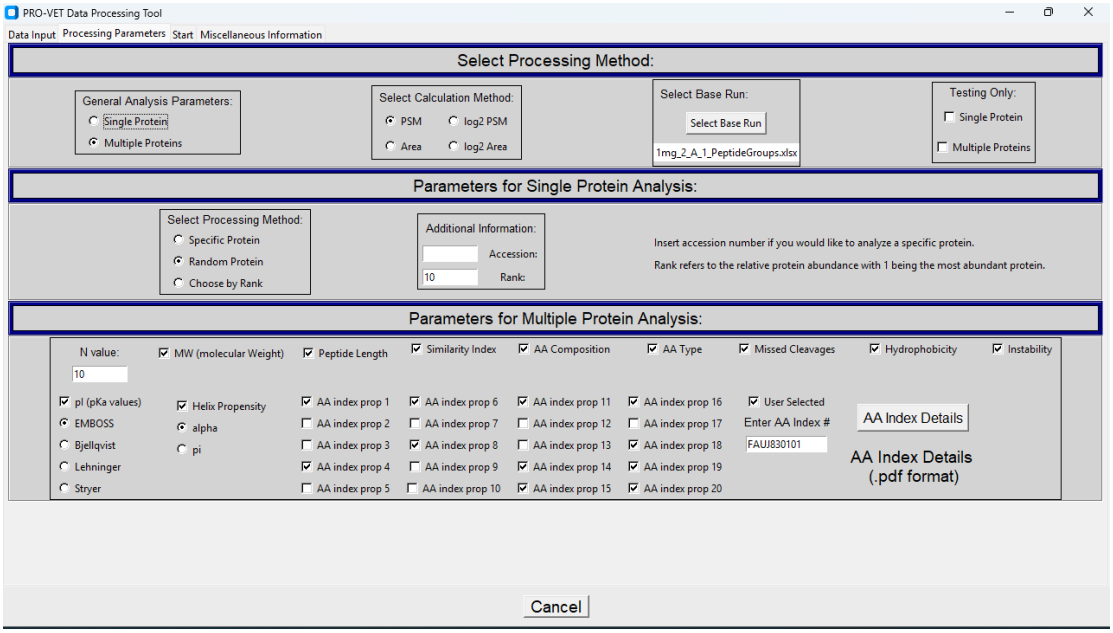
- **Additional Information.** Add the need information, when needed. (See Protein Selection Method)

c. Parameters for Multiple Protein Analysis. (Only used for multiple protein Analysis.)

- **N-Value.** Select the number of proteins to include in the analysis. These will use the most abundant proteins based on overall spectral count. For example, N-Value of 10 (default) will examine peptides from the 10 most abundant proteins. For some analysis (e.g. Similarity Index), it is best to avoid proteins with a low spectral count. Increasing the N-value will also require more processing time and computer resources.
- **Peptide Properties.** Select the peptide properties to examine. These include:
    a. Peptide Molecular Weight (MW). Monoisotopic Molecular Weight of peptides.
    b. Peptide Length. Number of Amino acids in the peptide.
    c. Similarity Index. (See manuscript).
    d. AA Composition. (Prevalence of each Amino acid, weighed by number of PSMs)
    e. AA Type. AAs can be classified in a number of different ways. For this study, the 7 categories used are i) aliphatic (A, G, I, L, P, V), aromatic (F, W, Y), sulfur containing (C, M), Hydroxyl (S, T), basic (R, H, K), acidic (D, E), and amide (N, Q).
    f. Number of Missed Cleavages. Missed Cleavages assumes the use of trypsin as the protease and cleavage of the peptide bonds on the C-terminal of lysine (K) and arginine (R) residues, but not before proline (Pro).
    g. Hydrophobicity (or gravy score). The Grand average of hydropathicity (GRAVY) of the linear polypeptide sequence is calculated as the sum of hydropathy values of all amino acids, divided by the number of residues in the sequence. Increasing positive score indicates greater hydrophobicity, but no account is taken of the way the protein folds in three dimensions or the percentage of residues buried in the hydrophobic core of the protein. The calculation is based on the Kyte-Doolittle scale [1].
    h. Instability Index. Based on the instability of presence of certain dipeptide sequences. Used the method of Guruprasad, *et al.* [2] as described by Gasteiger, *et al*. [3].
    i. Isoelectric Point (pI). The method described by Kozlowski [4] was used with an option of 4 different pKa values from EMBOSS [5], Lehninger [6], Styer [7], and Bjellqvist [8].
    j. Helix Propensity. Propensity of peptide to for either alpha- or pi- helix.

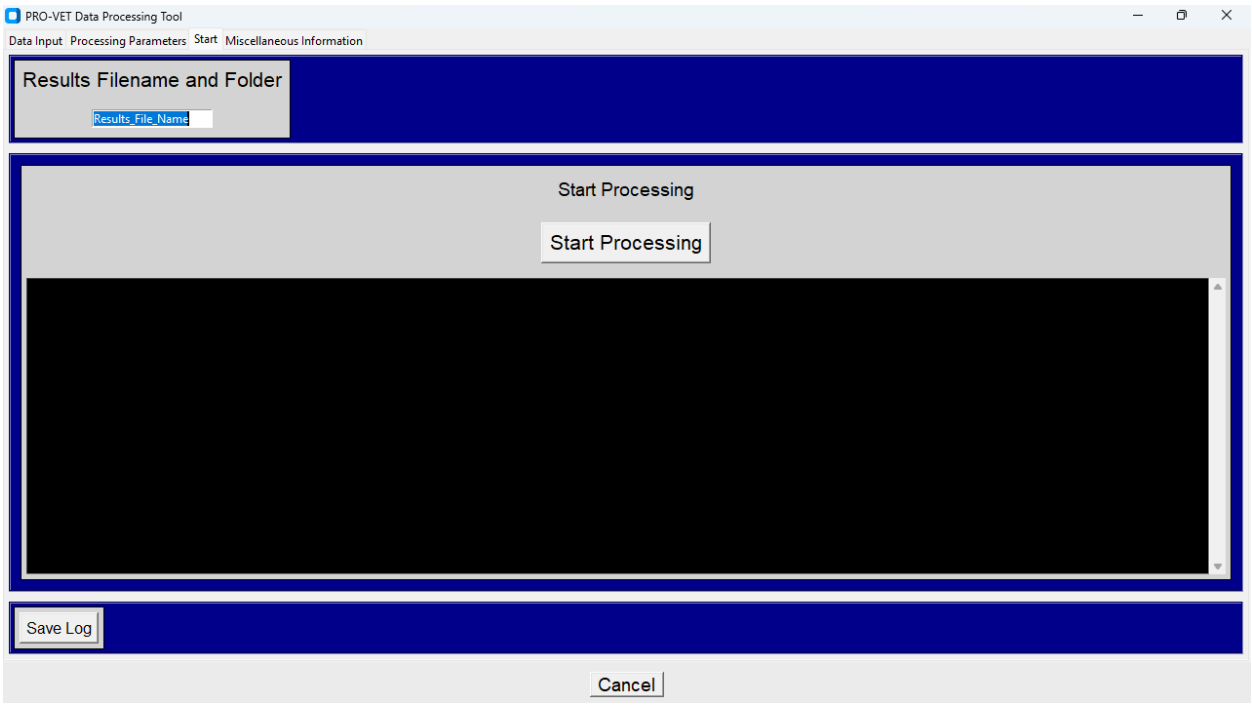
k. AA index properties. Twenty (20) are incorporated automatically. Any other can be specified (see table below).

The Parameter Selection Tab:



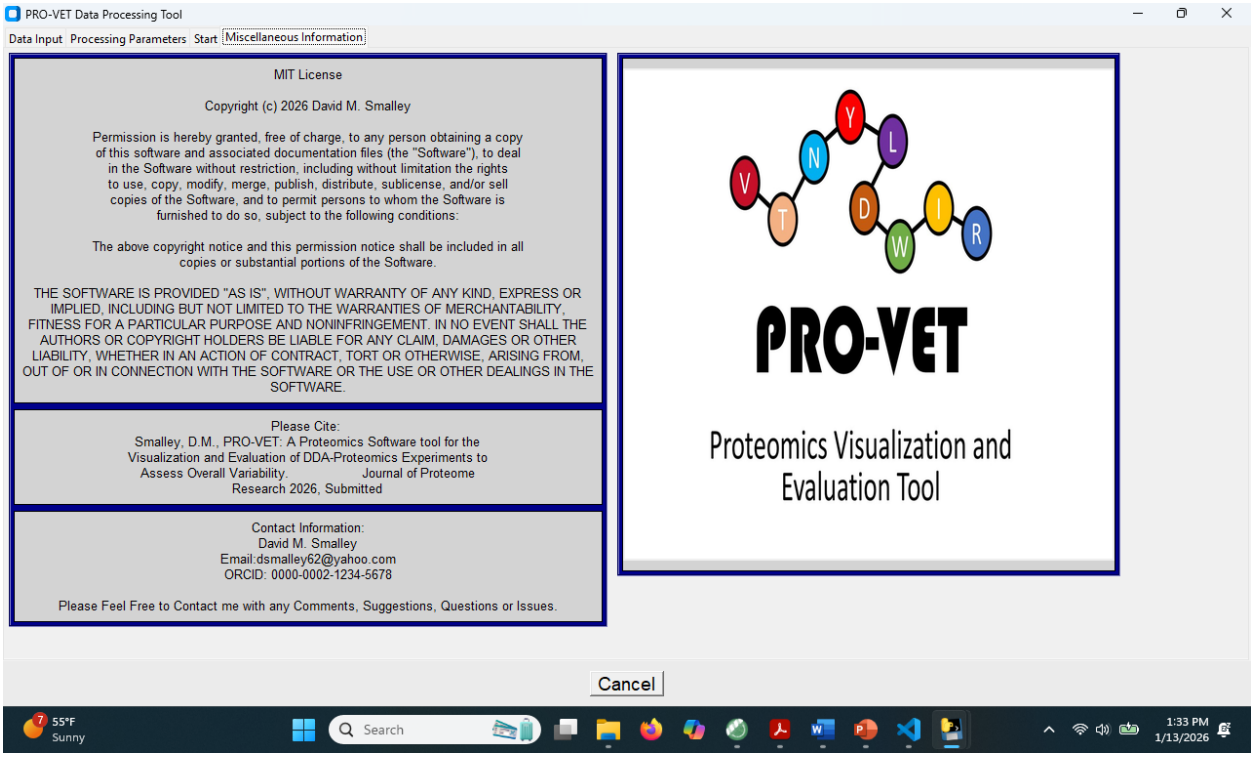
4) Start Tab.

- **Results Filename and Folder.** All results including data tables and figures will be placed I the indicated folder in the same folder where the Pro-VET file is located. These images should be better than those generated on-screen. Higher quality figures can be created using the created data files. All dataframes and figures are automatically saved following creation.
- **Start Processing.** Make sure all other parameters are properly selected prior to clicking the “Start Processing” file. Once clicked, it may take several minutes to generate results. Most of the computer run time is used to upload data. It is highly recommended that one examines a few files samples initially to become familiar with the program. Several sample files are provided.
- **Save Log.** If problems develop in sample processing, the log can be saved and it will help in troubleshooting.



5) Miscellaneous Information Tab.

Self Explanatory.



Importing Data from Search Engines.

General Comments: Run protein search engine and obtain results. Each raw file (.raw) should be searched independently. Data from most common search engines can be used but some may need to be manually manipulate prior to generate data that can be imported into Pro-Vet. There are others that can be exported or created by search engines and used directly. In addition, at least one fasta file is required with a ".fasta" extension. There are many formats possible for fasta files. This program works best for the fasta format used by uniport (www.uniprot.org). Other fasta formats should be tested prior to use.

Possible methods to import search results data:

a. Manually Generated Files. For the search results of each raw file, two **Comma-Separated Values** (.csv) files need to be created representing the proteins and peptides detected. The protein file should have at least four (4) columns for i) protein (column name = "description"), ii) accession number (column name = "accession") used during the spectral match search, iii) the number of spectral matches, also referred to as peptide spectral matches (column name = "PSM"), and iv) protein length (column name = "prot_len"). The protein files should be names with any identifier followed by an underscore ("_") and "proteins". For example, sample 1, would have a file names of "sample_1_proteins.csv".
Peptide files should also have at least four (4) columns. These are for i) sequence (column = "seq"), ii) protein accession number (column name = "accession"), iii) spectral count ("PSM"), and ion intensity ("intensity") of the peptide. If no intensity information is available, please make the intensity equal to 1 for all peptides. The peptide files should be names with any identifier followed by an underscore ("_") and "peptide". For example, sample 1, would have a file names of "sample_1_peptides.csv".

All these column and file names are case sensitive and do NOT include the quotation marks. These files should be relatively easy to generate in Excel for those familiar with that software and then exported as a .csv file. Sample data (ex. manual_input_example_peptides.csv and manual_input_example_1_protiens.csv) can be downloaded from the github page. (www.github.com/dsmalle62/Pro_VET/data)

b. Maxquant is a proteomics software package developed and updated by Jurgen Cox, Matthias Mann and coworkers for more than a decade [9,10]. It uses the Andromea search engine, is in widespread use, and is freely available. The version used to generate the test data was 2.7.3.0. Pro-Vet has been tested and works with this version. Each file search generates a "peptides.txt" and "proteinGroups.txt". These are typically in the "combined/txt" subdirectory and can be used directly by simply adding a sample identifier followed by an underscore ('_') prior to these. For example, sample 1 would create "sample_1_peptides.txt" and sample_1_proteinGroups.txt".
c. Proteome Discoverer™ is also a widely used proteomics software and is available from Thermo Fisher Scientific. While it is easy to use and vendor supported, it is not freely available and different versions and add-ons are available. The version utilized in the development of this software was 2.5.0.400 with Amanda 2.0 search Engine [11] using IMP-apQuant [12] for peptide quantitation. The Proteins and the Peptide Groups worksheets are exported as Excel workbooks and labeled with sample designation followed by "_Proteins" and "_ProteinGroups", respectively. These should have ".xlsx" extensions. For example, sample 1 data could be denoted as "sample_1_Proteins.xlsx" and "sample_1_PeptideGroups.xlsx".
d. MSFragger search engine and the FragPipe software were created by Nesvizhskii and coworkers [13]. This is also a nice proteomics software suite free to academic users. The version used and tested for this software was generated in version 23.1 of FragPipe. Each file generates a protein.tsv and a peptide.tsv. These tab-separated values (.tsv) files are similar to the .csv files. These files can also be used directly by adding an identifier followed by an underscore ("_") and then protein.tsv and peptide.tsv for the protein and peptide file, respectively. For example, sample 1 would be sample_1_protein.tsv and sample_2_peptide.tsv.

References:

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Appendix 2. AA Index. (See AA index [14] and (www.genome.jp/aaindex/) for details.

My code	Description	Reference	Citation
AA_index_001	alpha-CH chemical shifts	ANDN920101	Andersen et al., 1992
AA_index_002	Membrane-buried preference parameters	ARGP820103	Argos et al., 1982
AA_index_003	Residue volume	BIGC670101	Bigelow, 1967
AA_index_004	Retention coefficient in TFA	BROC820101	Browne et al., 1982
AA_index_005	Transfer free energy to surface	BULH740101	Bull-Breese, 1974
AA_index_006	Free energy of solution in water, kcal/mole	CHAM820102	Charton-Charton, 1982
AA_index_007	Entropy of formation	HUTJ700103	Hutchens, 1970
AA_index_008	Average accessible surface area	JANJ780101	Janin et al., 1978
AA_index_009	Average relative probability of beta-sheet	KANM800102	Kanehisa-Tsong, 1980
AA_index_010	Average relative probability of inner helix	KANM800103	Kanehisa-Tsong, 1980
AA_index_011	Zimm-Bragg parameter s at 20 C	SUEM840101	Sueki et al., 1984
AA_index_012	Electron-ion interaction potential	VELV850101	Veljkovic et al., 1985
AA_index_013	Unfolding Gibbs energy in water, pH7.0	YUTK870101	Yutani et al., 1987
AA_index_014	Amphiphilicity index	MITS020101	Mitaku et al., 2002
AA_index_015	Retention coefficient at pH 2	GUOD860101	Guo et al., 1986
AA_index_016	Buriability	ZHOH040103	Zhou-Zhou, 2004
AA_index_017	Average internal preferences	OLSK800101	Olsen, 1980
AA_index_018	Average eccentricity	KARS160105	Karkbara-Knisley, 2016
AA_index_019	Apparent partition energies calculated from Wertz-Scheraga index	GUYH850102	Guy,1985
AA_index_020	Polarity	ZIMJ680103	Zimmerman et al., 1968
	Hydrophobicity index	ARGP820101	Argos et al., 1982
	Signal sequence helical potential	ARGP820102	Argos et al., 1982
	Conformational parameter of inner helix	BEGF750101	Beghin-Dirkx, 1975
	Conformational parameter of beta-structure	BEGF750102	Beghin-Dirkx, 1975
	Conformational parameter of beta-turn	BEGF750103	Beghin-Dirkx, 1975
	Average flexibility indices	BHAR880101	Bhaskaran-Ponnuswamy, 1988
	Information value for accessibility; average fraction 35%	BIOV880101	Biou et al., 1998
	Information value for accessibility; average fraction 23%	BIOV880102	Biou et al., 1998

	Retention coefficient in HFBA	BROC820102	Browne et al., 1982
	Apparent partial specific volume	BULH740102	Bull-Breese, 1974
	alpha-NH chemical shifts	BUNA790101	Bundi-Wuthrich, 1979
	alpha-CH chemical shifts	BUNA790102	Bundi-Wuthrich, 1979
	Spin-spin coupling constants 3JHalpha-NH	BUNA790103	Bundi-Wuthrich, 1979
	Normalized frequency of alpha-helix	BURA740101	Burgess et al., 1974
	Normalized frequency of extended structure	BURA740102	Burgess et al., 1974
	Steric parameter	CHAM810101	Charton, 1981
	Polarizability parameter	CHAM820101	Charton-Charton, 1982
	The Chou-Fasman parameter of the coil conformation	CHAM830101	Charton-Charton, 1983
	A parameter defined from the residuals obtained from the best correlation of the Chou-Fasman parameter of beta-sheet	CHAM830102	Charton-Charton, 1983
	The number of atoms in the side chain labelled 1+1	CHAM830103	Charton-Charton, 1983
	The number of atoms in the side chain labelled 2+1	CHAM830104	Charton-Charton, 1983
	The number of atoms in the side chain labelled 3+1	CHAM830105	Charton-Charton, 1983
	The number of bonds in the longest chain	CHAM830106	Charton-Charton, 1983
	A parameter of charge transfer capability	CHAM830107	Charton-Charton, 1983
	A parameter of charge transfer donor capability	CHAM830108	Charton-Charton, 1983
	Average volume of buried residue	CHOC750101	Chothia, 1975
	Residue accessible surface area in tripeptide	CHOC760101	Chothia, 1976
	Residue accessible surface area in folded protein	CHOC760102	Chothia, 1976
	Proportion of residues 95% buried	CHOC760103	Chothia, 1976
	Proportion of residues 100% buried	CHOC760104	Chothia, 1976
	Normalized frequency of beta-turn	CHOP780101	Chou-Fasman, 1978a
	Normalized frequency of alpha-helix	CHOP780201	Chou-Fasman, 1978b
	Normalized frequency of beta-sheet	CHOP780202	Chou-Fasman, 1978b
	Normalized frequency of beta-turn	CHOP780203	Chou-Fasman, 1978b
	Normalized frequency of N-terminal helix	CHOP780204	Chou-Fasman, 1978b
	Normalized frequency of C-terminal helix	CHOP780205	Chou-Fasman, 1978b

	Normalized frequency of N-terminal non helical region	CHOP780206	Chou-Fasman, 1978b
	Normalized frequency of C-terminal non helical region	CHOP780207	Chou-Fasman, 1978b
	Normalized frequency of N-terminal beta-sheet	CHOP780208	Chou-Fasman, 1978b
	Normalized frequency of C-terminal beta-sheet	CHOP780209	Chou-Fasman, 1978b
	Normalized frequency of N-terminal non beta region	CHOP780210	Chou-Fasman, 1978b
	Normalized frequency of C-terminal non beta region	CHOP780211	Chou-Fasman, 1978b
	Frequency of the 1st residue in turn	CHOP780212	Chou-Fasman, 1978b
	Frequency of the 2nd residue in turn	CHOP780213	Chou-Fasman, 1978b
	Frequency of the 3rd residue in turn	CHOP780214	Chou-Fasman, 1978b
	Frequency of the 4th residue in turn	CHOP780215	Chou-Fasman, 1978b
	Normalized frequency of the 2nd and 3rd residues in turn	CHOP780216	Chou-Fasman, 1978b
	Normalized hydrophobicity scales for alpha-proteins	CIDH920101	Cid et al., 1992
	Normalized hydrophobicity scales for beta-proteins	CIDH920102	Cid et al., 1992
	Normalized hydrophobicity scales for alpha+beta-proteins	CIDH920103	Cid et al., 1992
	Normalized hydrophobicity scales for alpha/beta-proteins	CIDH920104	Cid et al., 1992
	Normalized average hydrophobicity scales	CIDH920105	Cid et al., 1992
	Partial specific volume	COHE430101	Cohn-Edsall, 1943
	Normalized frequency of middle helix	CRAJ730101	Crawford et al., 1973
	Normalized frequency of beta-sheet	CRAJ730102	Crawford et al., 1973
	Normalized frequency of turn	CRAJ730103	Crawford et al., 1973
	Size	DAWD720101	Dawson, 1972
	Amino acid composition	DAYM780101	Dayhoff et al., 1978a
	Relative mutability	DAYM780201	Dayhoff et al., 1978b
	Membrane preference for cytochrome b: MPH89	DESM900101	Degli Esposti et al., 1990
	Average membrane preference: AMP07	DESM900102	Degli Esposti et al., 1990
	Consensus normalized hydrophobicity scale	EISD840101	Eisenberg, 1984
	Solvation free energy	EISD860101	Eisenberg-McLachlan, 1986
	Atom-based hydrophobic moment	EISD860102	Eisenberg-McLachlan, 1986

	Direction of hydrophobic moment	EISD860103	Eisenberg-McLachlan, 1986
	Molecular weight	FASG760101	Fasman, 1976
	Melting point	FASG760102	Fasman, 1976
	Optical rotation	FASG760103	Fasman, 1976
	pK-N	FASG760104	Fasman, 1976
	pK-C	FASG760105	Fasman, 1976
	Hydrophobic parameter pi	FAUJ830101	Fauchere-Pliska, 1983
	Graph shape index	FAUJ880101	Fauchere et al., 1988
	Smoothed upsilon steric parameter	FAUJ880102	Fauchere et al., 1988
	Normalized van der Waals volume	FAUJ880103	Fauchere et al., 1988
	STERIMOL length of the side chain	FAUJ880104	Fauchere et al., 1988
	STERIMOL minimum width of the side chain	FAUJ880105	Fauchere et al., 1988
	STERIMOL maximum width of the side chain	FAUJ880106	Fauchere et al., 1988
	N.m.r. chemical shift of alpha-carbon	FAUJ880107	Fauchere et al., 1988
	Localized electrical effect	FAUJ880108	Fauchere et al., 1988
	Number of hydrogen bond donors	FAUJ880109	Fauchere et al., 1988
	Number of full nonbonding orbitals	FAUJ880110	Fauchere et al., 1988
	Positive charge	FAUJ880111	Fauchere et al., 1988
	Negative charge	FAUJ880112	Fauchere et al., 1988
	pK-a(RCOOH	FAUJ880113	Fauchere et al., 1988
	Helix-coil equilibrium constant	FINA770101	Finkelstein-Ptitsyn, 1977
	Helix initiation parameter at posision i-1	FINA910101	Finkelstein et al., 1991
	Helix initiation parameter at posision i,i+1,i+2	FINA910102	Finkelstein et al., 1991
	Helix termination parameter at posision j-2,j-1,j	FINA910103	Finkelstein et al., 1991
	Helix termination parameter at posision j+1	FINA910104	Finkelstein et al., 1991
	Partition coefficient	GARJ730101	Garel et al., 1973
	Alpha-helix indices	GEIM800101	Geisow-Roberts, 1980
	Alpha-helix indices for alpha-proteins	GEIM800102	Geisow-Roberts, 1980

	Alpha-helix indices for beta-proteins	GEIM800103	Geisow-Roberts, 1980
	Alpha-helix indices for alpha/beta-proteins	GEIM800104	Geisow-Roberts, 1980
	Beta-strand indices	GEIM800105	Geisow-Roberts, 1980
	Beta-strand indices for beta-proteins	GEIM800106	Geisow-Roberts, 1980
	Beta-strand indices for alpha/beta-proteins	GEIM800107	Geisow-Roberts, 1980
	Aperiodic indices	GEIM800108	Geisow-Roberts, 1980
	Aperiodic indices for alpha-proteins	GEIM800109	Geisow-Roberts, 1980
	Aperiodic indices for beta-proteins	GEIM800110	Geisow-Roberts, 1980
	Aperiodic indices for alpha/beta-proteins	GEIM800111	Geisow-Roberts, 1980
	Hydrophobicity factor	GOLD730101	Goldsack-Chalifoux, 1973
	Residue volume	GOLD730102	Goldsack-Chalifoux, 1973
	Composition	GRAR740101	Grantham, 1974
	Polarity	GRAR740102	Grantham, 1974
	Volume	GRAR740103	Grantham, 1974
	Partition energy	GUYH850101	Guy, 1985
	Hydration number	HOPA770101	Hopfinger, 1971
	Hydrophilicity value	HOPT810101	Hopp-Woods, 1981
	Heat capacity	HUTJ700101	Hutchens, 1970
	Absolute entropy	HUTJ700102	Hutchens, 1970
	Normalized relative frequency of alpha-helix	ISOY800101	Isogai et al., 1980
	Normalized relative frequency of extended structure	ISOY800102	Isogai et al., 1980
	Normalized relative frequency of bend	ISOY800103	Isogai et al., 1980
	Normalized relative frequency of bend R	ISOY800104	Isogai et al., 1980
	Normalized relative frequency of bend S	ISOY800105	Isogai et al., 1980
	Normalized relative frequency of helix end	ISOY800106	Isogai et al., 1980
	Normalized relative frequency of double bend	ISOY800107	Isogai et al., 1980
	Normalized relative frequency of coil	ISOY800108	Isogai et al., 1980
	Percentage of buried residues	JANJ780102	Janin et al., 1978
	Percentage of exposed residues	JANJ780103	Janin et al., 1978
	Ratio of buried and accessible molar fractions	JANJ790101	Janin, 1979
	Transfer free energy	JANJ790102	Janin, 1979
	Hydrophobicity	JOND750101	Jones, 1975
	pK (-COOH)	JOND750102	Jones, 1975
	Relative frequency of occurrence	JOND920101	Jones et al., 1992

	Relative mutability	JOND920102	Jones et al., 1992
	Amino acid distribution	JUKT750101	Jukes et al., 1975
	Sequence frequency	JUNJ780101	Jungck, 1978
	Average relative probability of helix	KANM800101	Kanehisa-Tsong, 1980
	Average relative probability of inner beta-sheet	KANM800104	Kanehisa-Tsong, 1980
	Flexibility parameter for no rigid neighbors	KARP850101	Karplus-Schulz, 1985
	Flexibility parameter for one rigid neighbor	KARP850102	Karplus-Schulz, 1985
	Flexibility parameter for two rigid neighbors	KARP850103	Karplus-Schulz, 1985
	The Kerr-constant increments	KHAG800101	Khanarian-Moore, 1980
	Net charge	KLEP840101	Klein et al., 1984
	Side chain interaction parameter	KRIW710101	Krigbaum-Rubin, 1971
	Side chain interaction parameter	KRIW790101	Krigbaum-Komoriya, 1979
	Fraction of site occupied by water	KRIW790102	Krigbaum-Komoriya, 1979
	Side chain volume	KRIW790103	Krigbaum-Komoriya, 1979
	Hydropathy index	KYTJ820101	Kyte-Doolittle, 1982
	Transfer free energy, CHP/water	LAWE840101	Lawson et al., 1984
	Hydrophobic parameter	LEVM760101	Levitt, 1976
	Distance between C-alpha and centroid of side chain	LEVM760102	Levitt, 1976
	Side chain angle theta(AAR)	LEVM760103	Levitt, 1976
	Side chain torsion angle phi	LEVM760104	Levitt, 1976
	Radius of gyration of side chain	LEVM760105	Levitt, 1976
	van der Waals parameter R0	LEVM760106	Levitt, 1976
	van der Waals parameter epsilon	LEVM760107	Levitt, 1976
	Normalized frequency of alpha-helix, with weights	LEVM780101	Levitt, 1978
	Normalized frequency of beta-sheet, with weights	LEVM780102	Levitt, 1978
	Normalized frequency of reverse turn, with weights	LEVM780103	Levitt, 1978
	Normalized frequency of alpha-helix, unweighted	LEVM780104	Levitt, 1978
	Normalized frequency of beta-sheet, unweighted	LEVM780105	Levitt, 1978
	Normalized frequency of reverse turn, unweighted	LEVM780106	Levitt, 1978
	Frequency of occurrence in beta-bends	LEWP710101	Lewis et al., 1971
	Conformational preference for all beta-strands	LIFS790101	Lifson-Sander, 1979
	Conformational preference for parallel beta-strands	LIFS790102	Lifson-Sander, 1979

	Conformational preference for antiparallel beta-strands	LIFS790103	Lifson-Sander, 1979
	Average surrounding hydrophobicity	MANP780101	Manavalan-Ponnuswamy, 1978
	Normalized frequency of alpha-helix	MAXF760101	Maxfield-Scheraga, 1976
	Normalized frequency of extended structure	MAXF760102	Maxfield-Scheraga, 1976
	Normalized frequency of zeta R	MAXF760103	Maxfield-Scheraga, 1976
	Normalized frequency of left-handed alpha-helix	MAXF760104	Maxfield-Scheraga, 1976
	Normalized frequency of zeta L	MAXF760105	Maxfield-Scheraga, 1976
	Normalized frequency of alpha region	MAXF760106	Maxfield-Scheraga, 1976
	Refractivity	MCMT640101	McMeekin et al., 1964
	Retention coefficient in HPLC, pH7.4	MEEJ800101	Meek, 1980
	Retention coefficient in HPLC, pH2.1	MEEJ800102	Meek, 1980
	Retention coefficient in NaClO4	MEEJ810101	Meek-Rossetti, 1981
	Retention coefficient in NaH2PO4	MEEJ810102	Meek-Rossetti, 1981
	Average reduced distance for C-alpha	MEIH800101	Meirovitch et al., 1980
	Average reduced distance for side chain	MEIH800102	Meirovitch et al., 1980
	Average side chain orientation angle	MEIH800103	Meirovitch et al., 1980
	Effective partition energy	MIYS850101	Miyazawa-Jernigan, 1985
	Normalized frequency of alpha-helix	NAGK730101	Nagano, 1973
	Normalized frequency of bata-structure	NAGK730102	Nagano, 1973
	Normalized frequency of coil	NAGK730103	Nagano, 1973
	AA composition of total proteins	NAKH900101	Nakashima et al., 1990
	SD of AA composition of total proteins	NAKH900102	Nakashima et al., 1990
	AA composition of mt-proteins	NAKH900103	Nakashima et al., 1990
	Normalized composition of mt-proteins	NAKH900104	Nakashima et al., 1990
	AA composition of mt-proteins from animal	NAKH900105	Nakashima et al., 1990
	Normalized composition from animal	NAKH900106	Nakashima et al., 1990
	AA composition of mt-proteins from fungi and plant	NAKH900107	Nakashima et al., 1990

	Normalized composition from fungi and plant	NAKH900108	Nakashima et al., 1990
	AA composition of membrane proteins	NAKH900109	Nakashima et al., 1990
	Normalized composition of membrane proteins	NAKH900110	Nakashima et al., 1990
	Transmembrane regions of non-mt-proteins	NAKH900111	Nakashima et al., 1990
	Transmembrane regions of mt-proteins	NAKH900112	Nakashima et al., 1990
	Ratio of average and computed composition	NAKH900113	Nakashima et al., 1990
	AA composition of CYT of single-spanning proteins	NAKH920101	Nakashima-Nishikawa, 1992
	AA composition of CYT2 of single-spanning proteins	NAKH920102	Nakashima-Nishikawa, 1992
	AA composition of EXT of single-spanning proteins	NAKH920103	Nakashima-Nishikawa, 1992
	AA composition of EXT2 of single-spanning proteins	NAKH920104	Nakashima-Nishikawa, 1992
	AA composition of MEM of single-spanning proteins	NAKH920105	Nakashima-Nishikawa, 1992
	AA composition of CYT of multi-spanning proteins	NAKH920106	Nakashima-Nishikawa, 1992
	AA composition of EXT of multi-spanning proteins	NAKH920107	Nakashima-Nishikawa, 1992
	AA composition of MEM of multi-spanning proteins	NAKH920108	Nakashima-Nishikawa, 1992
	8 A contact number	NISK800101	Nishikawa-Ooi, 1980
	14 A contact number	NISK860101	Nishikawa-Ooi, 1986
	Transfer energy, organic solvent/water	NOZY710101	Nozaki-Tanford, 1971
	Average non-bonded energy per atom	OOBM770101	Oobatake-Ooi, 1977
	Short and medium range non-bonded energy per atom	OOBM770102	Oobatake-Ooi, 1977
	Long range non-bonded energy per atom	OOBM770103	Oobatake-Ooi, 1977
	Average non-bonded energy per residue	OOBM770104	Oobatake-Ooi, 1977
	Short and medium range non-bonded energy per residue	OOBM770105	Oobatake-Ooi, 1977
	Optimized beta-structure-coil equilibrium constant	OOBM850101	Oobatake et al., 1985
	Optimized propensity to form reverse turn	OOBM850102	Oobatake et al., 1985
	Optimized transfer energy parameter	OOBM850103	Oobatake et al., 1985
	Optimized average non-bonded energy per atom	OOBM850104	Oobatake et al., 1985

	Optimized side chain interaction parameter	OOBM850105	Oobatake et al., 1985
	Normalized frequency of alpha-helix from LG	PALJ810101	Palau et al., 1981
	Normalized frequency of alpha-helix from CF	PALJ810102	Palau et al., 1981
	Normalized frequency of beta-sheet from LG	PALJ810103	Palau et al., 1981
	Normalized frequency of beta-sheet from CF	PALJ810104	Palau et al., 1981
	Normalized frequency of turn from LG	PALJ810105	Palau et al., 1981
	Normalized frequency of turn from CF	PALJ810106	Palau et al., 1981
	Normalized frequency of alpha-helix in all-alpha class	PALJ810107	Palau et al., 1981
	Normalized frequency of alpha-helix in alpha+beta class	PALJ810108	Palau et al.,1981
	Normalized frequency of alpha-helix in alpha/beta class	PALJ810109	Palau et al.,1981
	Normalized frequency of beta-sheet in all-beta class	PALJ810110	Palau et al., 1981
	Normalized frequency of beta-sheet in alpha+beta class	PALJ810111	Palau et al., 1981
	Normalized frequency of beta-sheet in alpha/beta class	PALJ810112	Palau et al., 1981
	Normalized frequency of turn in all-alpha class	PALJ810113	Palau et al., 1981
	Normalized frequency of turn in all-beta class	PALJ810114	Palau et al., 1981
	Normalized frequency of turn in alpha+beta class	PALJ810115	Palau et al., 1981
	Normalized frequency of turn in alpha/beta class	PALJ810116	Palau et al., 1981
	HPLC parameter	PARJ860101	Parker et al., 1986
	Partition coefficient	PLIV810101	Pliska et al., 1981
	Surrounding hydrophobicity in folded form	PONP800101	Ponnuswamy et al., 1980
	Average gain in surrounding hydrophobicity	PONP800102	Ponnuswamy et al., 1980
	Average gain ratio in surrounding hydrophobicity	PONP800103	Ponnuswamy et al., 1980
	Surrounding hydrophobicity in alpha-helix	PONP800104	Ponnuswamy et al., 1980
	Surrounding hydrophobicity in beta-sheet	PONP800105	Ponnuswamy et al., 1980
	Surrounding hydrophobicity in turn	PONP800106	Ponnuswamy et al., 1980
	Accessibility reduction ratio	PONP800107	Ponnuswamy et al., 1980
	Average number of surrounding residues	PONP800108	Ponnuswamy et al., 1980
	Intercept in regression analysis	PRAM820101	Prabhakaran-Ponnuswamy, 1982
	Slope in regression analysis x 1.0E1	PRAM820102	Prabhakaran-Ponnuswamy, 1982

	Correlation coefficient in regression analysis	PRAM820103	Prabhakaran-Ponnuswamy,1982
	Hydrophobicity	PRAM900101	Prabhakaran, 1990
	Relative frequency in alpha-helix	PRAM900102	Prabhakaran, 1990
	Relative frequency in beta-sheet	PRAM900103	Prabhakaran, 1990
	Relative frequency in reverse-turn	PRAM900104	Prabhakaran, 1990
	Helix-coil equilibrium constant	PTIO830101	Ptitsyn-Finkelstein, 1983
	Beta-coil equilibrium constant	PTIO830102	Ptitsyn-Finkelstein, 1983
	Weights for alpha-helix at the window position of -6	QIAN880101	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of -5	QIAN880102	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of -4	QIAN880103	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of -3	QIAN880104	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of -2	QIAN880105	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of -1	QIAN880106	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of 0	QIAN880107	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of 1	QIAN880108	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of 2	QIAN880109	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of 3	QIAN880110	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of 4	QIAN880111	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of 5	QIAN880112	Qian-Sejnowski, 1988
	Weights for alpha-helix at the window position of 6	QIAN880113	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of -6	QIAN880114	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of -5	QIAN880115	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of -4	QIAN880116	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of -3	QIAN880117	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of -2	QIAN880118	Qian-Sejnowski, 1988

	Weights for beta-sheet at the window position of -1	QIAN880119	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of 0	QIAN880120	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of 1	QIAN880121	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of 2	QIAN880122	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of 3	QIAN880123	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of 4	QIAN880124	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of 5	QIAN880125	Qian-Sejnowski, 1988
	Weights for beta-sheet at the window position of 6	QIAN880126	Qian-Sejnowski, 1988
	Weights for coil at the window position of -6	QIAN880127	Qian-Sejnowski, 1988
	Weights for coil at the window position of -5	QIAN880128	Qian-Sejnowski, 1988
	Weights for coil at the window position of -4	QIAN880129	Qian-Sejnowski, 1988
	Weights for coil at the window position of -3	QIAN880130	Qian-Sejnowski, 1988
	Weights for coil at the window position of -2	QIAN880131	Qian-Sejnowski, 1988
	Weights for coil at the window position of -1	QIAN880132	Qian-Sejnowski, 1988
	Weights for coil at the window position of 0	QIAN880133	Qian-Sejnowski, 1988
	Weights for coil at the window position of 1	QIAN880134	Qian-Sejnowski, 1988
	Weights for coil at the window position of 2	QIAN880135	Qian-Sejnowski, 1988
	Weights for coil at the window position of 3	QIAN880136	Qian-Sejnowski, 1988
	Weights for coil at the window position of 4	QIAN880137	Qian-Sejnowski, 1988
	Weights for coil at the window position of 5	QIAN880138	Qian-Sejnowski, 1988
	Weights for coil at the window position of 6	QIAN880139	Qian-Sejnowski, 1988
	Average reduced distance for C-alpha	RACS770101	Rackovsky-Scheraga, 1977
	Average reduced distance for side chain	RACS770102	Rackovsky-Scheraga, 1977
	Side chain orientational preference	RACS770103	Rackovsky-Scheraga, 1977
	Average relative fractional occurrence in A0(i)	RACS820101	Rackovsky-Scheraga, 1982
	Average relative fractional occurrence in AR(i) (Rackovsky-Scheraga, 1982)	RACS820102	Rackovsky-Scheraga, 1983

	Average relative fractional occurrence in AL(i)	RACS820103	Rackovsky-Scheraga, 1984
	Average relative fractional occurrence in EL(i)	RACS820104	Rackovsky-Scheraga, 1985
	Average relative fractional occurrence in E0(i)	RACS820105	Rackovsky-Scheraga, 1986
	Average relative fractional occurrence in ER(i)	RACS820106	Rackovsky-Scheraga, 1987
	Average relative fractional occurrence in A0(i-1)	RACS820107	Rackovsky-Scheraga, 1988
	Average relative fractional occurrence in AR(i-1)	RACS820108	Rackovsky-Scheraga, 1989
	Average relative fractional occurrence in AL(i-1)	RACS820109	Rackovsky-Scheraga, 1990
	Average relative fractional occurrence in EL(i-1)	RACS820110	Rackovsky-Scheraga, 1991
	Average relative fractional occurrence in E0(i-1)	RACS820111	Rackovsky-Scheraga, 1992
	Average relative fractional occurrence in ER(i-1)	RACS820112	Rackovsky-Scheraga, 1993
	Value of theta(i)	RACS820113	Rackovsky-Scheraga, 1994
	Value of theta(i-1)	RACS820114	Rackovsky-Scheraga, 1995
	Transfer free energy from chx to wat	RADA880101	Radzicka-Wolfenden, 1988
	Transfer free energy from oct to wat	RADA880102	Radzicka-Wolfenden, 1988
	Transfer free energy from vap to chx	RADA880103	Radzicka-Wolfenden, 1988
	Transfer free energy from chx to oct	RADA880104	Radzicka-Wolfenden, 1988
	Transfer free energy from vap to oct	RADA880105	Radzicka-Wolfenden, 1988
	Accessible surface area	RADA880106	Radzicka-Wolfenden, 1988
	Energy transfer from out to in(95%buried)	RADA880107	95%buried
	Mean polarity	RADA880108	Radzicka-Wolfenden, 1988
	Relative preference value at N"	RICJ880101	Richardson-Richardson, 1988
	Relative preference value at N'	RICJ880102	Richardson-Richardson, 1988
	Relative preference value at N-cap	RICJ880103	Richardson-Richardson, 1988
	Relative preference value at N1	RICJ880104	Richardson-Richardson, 1988
	Relative preference value at N2	RICJ880105	Richardson-Richardson, 1988
	Relative preference value at N3	RICJ880106	Richardson-Richardson, 1988

	Relative preference value at N4	RICJ880107	Richardson-Richardson, 1988
	Relative preference value at N5	RICJ880108	Richardson-Richardson, 1988
	Relative preference value at Mid	RICJ880109	Richardson-Richardson, 1988
	Relative preference value at C5	RICJ880110	Richardson-Richardson, 1988
	Relative preference value at C4	RICJ880111	Richardson-Richardson, 1988
	Relative preference value at C3	RICJ880112	Richardson-Richardson, 1988
	Relative preference value at C2	RICJ880113	Richardson-Richardson, 1988
	Relative preference value at C1	RICJ880114	Richardson-Richardson, 1988
	Relative preference value at C-cap	RICJ880115	Richardson-Richardson, 1988
	Relative preference value at C'	RICJ880116	Richardson-Richardson, 1988
	Relative preference value at C"	RICJ880117	Richardson-Richardson, 1988
	Information measure for alpha-helix	ROBB760101	Robson-Suzuki, 1976
	Information measure for N-terminal helix	ROBB760102	Robson-Suzuki, 1976
	Information measure for middle helix	ROBB760103	Robson-Suzuki, 1976
	Information measure for C-terminal helix	ROBB760104	Robson-Suzuki, 1976
	Information measure for extended	ROBB760105	Robson-Suzuki, 1976
	Information measure for pleated-sheet	ROBB760106	Robson-Suzuki, 1976
	Information measure for extended without H-bond	ROBB760107	Robson-Suzuki, 1976
	Information measure for turn	ROBB760108	Robson-Suzuki, 1976
	Information measure for N-terminal turn	ROBB760109	Robson-Suzuki, 1976
	Information measure for middle turn	ROBB760110	Robson-Suzuki, 1976
	Information measure for C-terminal turn	ROBB760111	Robson-Suzuki, 1976
	Information measure for coil	ROBB760112	Robson-Suzuki, 1976
	Information measure for loop	ROBB760113	Robson-Suzuki, 1976
	Hydration free energy	ROBB790101	Robson-Osguthorpe, 1979
	Mean area buried on transfer	ROSG850101	Rose et al., 1985

	Mean fractional area loss	ROSG850102	Rose et al., 1985
	Side chain hydropathy, uncorrected for solvation	ROSM880101	Roseman, 1988
	Side chain hydropathy, corrected for solvation	ROSM880102	Roseman, 1988
	Loss of Side chain hydropathy by helix formation	ROSM880103	Roseman, 1988
	Transfer free energy	SIMZ760101	Simon, 1976
	Principal component I	SNEP660101	Sneath, 1966
	Principal component II	SNEP660102	Sneath, 1966
	Principal component III	SNEP660103	Sneath, 1966
	Principal component IV	SNEP660104	Sneath, 1966
	Zimm-Bragg parameter sigma x 1.0E4	SUEM840102	Sueki et al., 1984
	Optimal matching hydrophobicity	SWER830101	Sweet-Eisenberg, 1983
	Normalized frequency of alpha-helix	TANS770101	Tanaka-Scheraga, 1977
	Normalized frequency of isolated helix	TANS770102	Tanaka-Scheraga, 1977
	Normalized frequency of extended structure	TANS770103	Tanaka-Scheraga, 1977
	Normalized frequency of chain reversal R	TANS770104	Tanaka-Scheraga, 1977
	Normalized frequency of chain reversal S	TANS770105	Tanaka-Scheraga, 1977
	Normalized frequency of chain reversal D	TANS770106	Tanaka-Scheraga, 1977
	Normalized frequency of left-handed helix	TANS770107	Tanaka-Scheraga, 1977
	Normalized frequency of zeta R	TANS770108	Tanaka-Scheraga, 1977
	Normalized frequency of coil	TANS770109	Tanaka-Scheraga, 1977
	Normalized frequency of chain reversal	TANS770110	Tanaka-Scheraga, 1977
	Relative population of conformational state A	VASM830101	Vasquez et al., 1983
	Relative population of conformational state C	VASM830102	Vasquez et al., 1983
	Relative population of conformational state E	VASM830103	Vasquez et al., 1983
	Bitterness	VENT840101	Venanzi, 1984
	Transfer free energy to lipophilic phase	VHEG790101	von Heijne-Blomberg, 1979
	Average interactions per side chain atom	WARP780101	Warme-Morgan, 1978
	RF value in high salt chromatography	WEBA780101	Weber-Lacey, 1978
	Propensity to be buried inside	WERD780101	Wertz-Scheraga, 1978
	Free energy change of epsilon(i) to epsilon(ex)	WERD780102	Wertz-Scheraga, 1978

	Free energy change of alpha(Ri) to alpha(Rh)	WERD780103	Wertz-Scheraga, 1979
	Free energy change of epsilon(i) to alpha(Rh)	WERD780104	Wertz-Scheraga, 1980
	Polar requirement	WOEC730101	Woese, 1973
	Hydration potential	WOLR810101	Wolfenden et al., 1981
	Principal property value z1	WOLS870101	Wold et al., 1987
	Principal property value z2	WOLS870102	Wold et al., 1987
	Principal property value z3	WOLS870103	Wold et al., 1987
	Unfolding Gibbs energy in water, pH9.0	YUTK870102	Yutani et al., 1987
	Activation Gibbs energy of unfolding, pH7.0	YUTK870103	Yutani et al., 1987
	Activation Gibbs energy of unfolding, pH9.0	YUTK870104	Yutani et al., 1987
	Dependence of partition coefficient on ionic strength	ZASB820101	Zaslavsky et al., 1982
	Hydrophobicity	ZIMJ680101	Zimmerman et al., 1968
	Bulkiness	ZIMJ680102	Zimmerman et al., 1968
	Isoelectric point	ZIMJ680104	Zimmerman et al., 1968
	RF rank	ZIMJ680105	Zimmerman et al., 1968
	Normalized positional residue frequency at helix termini N4'	AURR980101	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini N"'	AURR980102	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini N"	AURR980103	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini N'	AURR980104	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini Nc	AURR980105	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini N1	AURR980106	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini N2	AURR980107	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini N3	AURR980108	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini N4	AURR980109	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini N5	AURR980110	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini C5	AURR980111	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini C4	AURR980112	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini C3	AURR980113	Aurora-Rose, 1998

	Normalized positional residue frequency at helix termini C2	AURR980114	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini C1	AURR980115	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini Cc	AURR980116	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini C'	AURR980117	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini C"	AURR980118	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini C"'	AURR980119	Aurora-Rose, 1998
	Normalized positional residue frequency at helix termini C4'	AURR980120	Aurora-Rose, 1998
	Delta G values for the peptides extrapolated to 0 M urea	ONEK900101	O'Neil-DeGrado, 1990
	Helix formation parameters (delta delta G)	ONEK900102	O'Neil-DeGrado, 1990
	Normalized flexibility parameters (B-values), average	VINM940101	Vihinen et al., 1994
	Normalized flexibility parameters (B-values) for each residue surrounded by none rigid neighbours	VINM940102	Vihinen et al., 1995
	Normalized flexibility parameters (B-values) for each residue surrounded by one rigid neighbours	VINM940103	Vihinen et al., 1996
	Normalized flexibility parameters (B-values) for each residue surrounded by two rigid neighbours	VINM940104	Vihinen et al., 1997
	Free energy in alpha-helical conformation	MUNV940101	Munoz-Serrano, 1994
	Free energy in alpha-helical region	MUNV940102	Munoz-Serrano, 1994
	Free energy in beta-strand conformation	MUNV940103	Munoz-Serrano, 1994
	Free energy in beta-strand region	MUNV940104	Munoz-Serrano, 1994
	Free energy in beta-strand region	MUNV940105	Munoz-Serrano, 1994
	Free energies of transfer of AcWl-X-LL peptides from bilayer interfaceto water	WIMW960101	Wimley-White, 1996
	Thermodynamic beta sheet propensity	KIMC930101	Kim-Berg, 1993
	Turn propensity scale for transmembrane helices	MONM990101	Monne et al., 1999
	Alpha helix propensity of position 44 in T4 lysozyme	BLAM930101	Blaber et al., 1993
	p-Values of mesophilic proteins based on the distributions of B values	PARS000101	Parthasarathy-Murthy, 2000
	p-Values of thermophilic proteins based on the distributions of B values	PARS000102	Parthasarathy-Murthy, 2000
	Distribution of amino acid residues in the 18 non-redundant families ofthermophilic proteins	KUMS000101	Kumar et al., 2000
	Distribution of amino acid residues in the 18 non-redundant families ofmesophilic proteins	KUMS000102	Kumar et al., 2000

	Distribution of amino acid residues in the alpha-helices in thermophilic proteins	KUMS000103	Kumar et al., 2000
	Distribution of amino acid residues in the alpha-helices in mesophilicproteins	KUMS000104	Kumar et al., 2000
	Side-chain contribution to protein stability (kJ/mol)	TAKK010101	Takano-Yutani, 2001
	Propensity of amino acids within pi-helices	FODM020101	Fodje-Al-Karadaghi, 2002
	Hydropathy scale based on self-information values in the two-state model (5% accessibility)	NADH010101	(Naderi-Manesh et al., 2001
	Hydropathy scale based on self-information values in the two-state model (9% accessibility)	NADH010102	(Naderi-Manesh et al., 2002
	Hydropathy scale based on self-information values in the two-state model (16% accessibility)	NADH010103	(Naderi-Manesh et al., 2003
	Hydropathy scale based on self-information values in the two-state model (20% accessibility)	NADH010104	(Naderi-Manesh et al., 2004
	Hydropathy scale based on self-information values in the two-state model (25% accessibility)	NADH010105	(Naderi-Manesh et al., 2005
	Hydropathy scale based on self-information values in the two-state model (36% accessibility)	NADH010106	(Naderi-Manesh et al., 2006
	Hydropathy scale based on self-information values in the two-state model (50% accessibility)	NADH010107	(Naderi-Manesh et al., 2007
	Averaged turn propensities in a transmembrane helix	MONM990201	Monne et al., 1999
	Alpha-helix propensity derived from designed sequences	KOEP990101	Koehl-Levitt, 1999
	Beta-sheet propensity derived from designed sequences	KOEP990102	Koehl-Levitt, 1999
	Composition of amino acids in extracellular proteins (%)	CEDJ970101	Cedanoet al., 1997
	Composition of amino acids in anchored proteins (%)	CEDJ970102	Cedanoet al., 1997
	Composition of amino acids in membrane proteins (%)	CEDJ970103	Cedanoet al., 1997
	Composition of amino acids in intracellular proteins (%)	CEDJ970104	Cedanoet al., 1997
	Composition of amino acids in nuclear proteins (%)	CEDJ970105	Cedanoet al., 1997
	Surface composition of amino acids in intracellular proteins of thermophiles (%)	FUKS010101	Fukuchi-Nishikawa, 2001
	Surface composition of amino acids in intracellular proteins of mesophiles (%)	FUKS010102	Fukuchi-Nishikawa, 2001
	Surface composition of amino acids in extracellular proteins of mesophiles (%)	FUKS010103	Fukuchi-Nishikawa, 2001
	Surface composition of amino acids in nuclear proteins (%)	FUKS010104	Fukuchi-Nishikawa, 2001
	Interior composition of amino acids in intracellular proteins of thermophiles (%)	FUKS010105	Fukuchi-Nishikawa, 2001
	Interior composition of amino acids in intracellular proteins of mesophiles (%)	FUKS010106	Fukuchi-Nishikawa, 2001
	Interior composition of amino acids in extracellular proteins of mesophiles (%)	FUKS010107	Fukuchi-Nishikawa, 2001

	Interior composition of amino acids in nuclear proteins (%)	FUKS010108	Fukuchi-Nishikawa, 2001
	Entire chain composition of amino acids in intracellular proteins of thermophiles (%)	FUKS010109	Fukuchi-Nishikawa, 2001
	Entire chain composition of amino acids in intracellular proteins of mesophiles (%)	FUKS010110	Fukuchi-Nishikawa, 2001
	Entire chain composition of amino acids in extracellular proteins of mesophiles (%)	FUKS010111	Fukuchi-Nishikawa, 2001
	Entire chain compositino of amino acids in nuclear proteins (%)	FUKS010112	Fukuchi-Nishikawa, 2001
	Screening coefficients gamma, local	AVBF000101	Avbelj, 2000
	Screening coefficients gamma, non-local	AVBF000102	Avbelj, 2000
	Slopes tripeptide, FDPB VFF neutral	AVBF000103	Avbelj, 2000
	Slopes tripeptides, LD VFF neutral	AVBF000104	Avbelj, 2000
	Slopes tripeptide, FDPB VFF noside	AVBF000105	Avbelj, 2000
	Slopes tripeptide FDPB VFF all	AVBF000106	Avbelj, 2000
	Slopes tripeptide FDPB PARSE neutral	AVBF000107	Avbelj, 2000
	Slopes dekapeptide, FDPB VFF neutral	AVBF000108	Avbelj, 2000
	Slopes proteins, FDPB VFF neutral	AVBF000109	Avbelj, 2000
	Side-chain conformation by gaussian evolutionary method	YANJ020101	Yang et al., 2002
	Volumes including the crystallographic waters using the ProtOr	TSAJ990101	Tsai etal., 1999
	Volumes not including the crystallographic waters using the ProtOr	TSAJ990102	Tsai et al., 1999
	Electron-ion interaction potential values	COSI940101	Cosic, 1994
	Hydrophobicity scales	PONP930101	Ponnuswamy, 1993
	Hydrophobicity coefficient in RP-HPLC, C18 with 0.1%TFA/MeCN/H2O	WILM950101	Wilceet al. 1995
	Hydrophobicity coefficient in RP-HPLC, C8 with 0.1%TFA/MeCN/H2O	WILM950102	Wilceet al. 1995
	Hydrophobicity coefficient in RP-HPLC, C4 with 0.1%TFA/MeCN/H2O	WILM950103	Wilceet al. 1995
	Hydrophobicity coefficient in RP-HPLC, C18 with 0.1%TFA/2-PrOH/MeCN/H2O	WILM950104	Wilce et al. 1995
	Hydrophilicity scale	KUHL950101	Kuhn et al., 1995
	Modified Kyte-Doolittle hydrophobicity scale	JURD980101	Juretic et al., 1998
	Interactivity scale obtained from the contact matrix	BASU050101	Bastolla et al.,2005
	Interactivity scale obtained by maximizing the mean of correlation coefficient over single-domain globular proteins	BASU050102	Bastolla et al., 2005
	Interactivity scale obtained by maximizing the mean of correlation coefficient over pairs of sequences sharing the TIM barrel fold	BASU050103	Bastolla et al., 2005
	Linker propensity index	SUYM030101	Suyama-Ohara, 2003

	Knowledge-based membrane-propensity scale from 1D_Helix in MPtopo databases	PUNT030101	Punta-Maritan, 2003
	Knowledge-based membrane-propensity scale from 3D_Helix in MPtopo databases	PUNT030102	Punta-Maritan, 2003
	Linker propensity from all dataset	GEOR030101	George-Heringa, 2003
	Linker propensity from 1-linker dataset	GEOR030102	George-Heringa, 2003
	Linker propensity from 2-linker dataset	GEOR030103	George-Heringa, 2003
	Linker propensity from 3-linker dataset	GEOR030104	George-Heringa, 2003
	Linker propensity from small dataset (linker length is less than six residues)	GEOR030105	(George-Heringa, 2003
	Linker propensity from medium dataset (linker length is between six and14 residues)	GEOR030106	(George-Heringa, 2003
	Linker propensity from long dataset (linker length is greater than 14residues)	GEOR030107	(George-Heringa, 2003
	Linker propensity from helical (annotated by DSSP) dataset	GEOR030108	(George-Heringa, 2003
	Linker propensity from non-helical (annotated by DSSP) dataset	GEOR030109	(George-Heringa, 2003
	The stability scale from the knowledge-based atom-atom potential	ZHOH040101	Zhou-Zhou, 2004
	The relative stability scale extracted from mutation experiments	ZHOH040102	Zhou-Zhou, 2004
	Linker index	BAEK050101	Bae et al., 2005
	Mean volumes of residues buried in protein interiors	HARY940101	Harpaz et al., 1994
	Average volumes of residues	PONJ960101	Pontius et al., 1996
	Hydrostatic pressure asymmetry index, PAI	DIGM050101	Di Giulio, 2005
	Hydrophobicity index	WOLR790101	Wolfenden et al., 1979
	Hydrophobicity-related index	KIDA850101	Kidera et al., 1985
	Apparent partition energies calculated from Robson-Osguthorpe index	GUYH850103	Guy, 1985
	Apparent partition energies calculated from Janin index	GUYH850104	Guy, 1985
	Apparent partition energies calculated from Chothia index	GUYH850105	Guy, 1985
	Hydropathies of amino acid side chains, neutral form	ROSM880104	Roseman, 1988
	Hydropathies of amino acid side chains, pi-values in pH 7.0	ROSM880105	Roseman, 1988
	Weights from the IFH scale	JACR890101	Jacobs-White, 1989
	Hydrophobicity index, 3.0 pH	COWR900101	Cowan-Whittaker, 1990
	Scaled side chain hydrophobicity values	BLAS910101	Black-Mould, 1991
	Hydrophobicity scale from native protein structures	CASG920101	Casari-Sippl, 1992

	NNEIG index	CORJ870101	Cornette et al., 1987
	SWEIG index	CORJ870102	Cornette et al., 1987
	PRIFT index	CORJ870103	Cornette et al., 1987
	PRILS index	CORJ870104	Cornette et al., 1987
	ALTFT index	CORJ870105	Cornette et al., 1987
	ALTLS index	CORJ870106	Cornette et al., 1987
	TOTFT index	CORJ870107	Cornette et al., 1987
	TOTLS index	CORJ870108	Cornette et al., 1987
	Relative partition energies derived by the Bethe approximation	MIYS990101	Miyazawa-Jernigan, 1999
	Optimized relative partition energies - method A	MIYS990102	Miyazawa-Jernigan, 1999
	Optimized relative partition energies - method B	MIYS990103	Miyazawa-Jernigan, 1999
	Optimized relative partition energies - method C	MIYS990104	Miyazawa-Jernigan, 1999
	Optimized relative partition energies - method D	MIYS990105	Miyazawa-Jernigan, 1999
	Hydrophobicity index	ENGD860101	Engelman et al., 1986
	Hydrophobicity index	FASG890101	Fasman, 1989
	Number of vertices (order of the graph)	KARS160101	Karkbara-Knisley, 2016
	Number of edges (size of the graph)	KARS160102	Karkbara-Knisley, 2016
	Total weighted degree of the graph (obtained by adding all the weightsof all the vertices)	KARS160103	Karkbara-Knisley, 2016
	Weighted domination number	KARS160104	Karkbara-Knisley, 2016
	Radius (minimum eccentricity)	KARS160106	Karkbara-Knisley, 2016
	Diameter (maximum eccentricity)	KARS160107	Karkbara-Knisley, 2016
	Average weighted degree (total degree, divided by the number of vertices	KARS160108	Karkbara-Knisley, 2016
	Maximum eigenvalue of the weighted Laplacian matrix of the graph	KARS160109	Karkbara-Knisley, 2016
	Minimum eigenvalue of the weighted Laplacian matrix of the graph	KARS160110	Karkbara-Knisley, 2016
	Average eigenvalue of the Laplacian matrix of the the graph	KARS160111	Karkbara-Knisley, 2016
	Second smallest eigenvalue of the Laplacian matrix of the graph	KARS160112	Karkbara-Knisley, 2016

	Weighted domination number using the atomic number	KARS160113	Karkbara-Knisley, 2016
	Average weighted eccentricity based on the the atomic number	KARS160114	Karkbara-Knisley, 2016
	Weighted radius based on the atomic number (minimum eccentricity)	KARS160115	Karkbara-Knisley, 2016
	Weighted diameter based on the atomic number (maximum eccentricity	KARS160116	Karkbara-Knisley, 2016
	Total weighted atomic number of the graph (obtained by summing all theatomic number of each of the vertices in the graph)	KARS160117	Karkbara-Knisley, 2016
	Average weighted atomic number or degree based on atomic number in thegraph	KARS160118	Karkbara-Knisley, 2016
	Weighted maximum eigenvalue based on the atomic numbers	KARS160119	Karkbara-Knisley, 2016
	Weighted minimum eigenvalue based on the atomic numbers	KARS160120	Karkbara-Knisley, 2016
	Weighted average eigenvalue based on the atomic numbers	KARS160121	Karkbara-Knisley, 2016
	Weighted second smallest eigenvalue of the weighted Laplacian matrix	KARS160122	Karkbara-Knisley, 2016